

DEEP LEARNING

MCQ Question Bank

Unit I – 20 Questions

Unit II – 20 Questions

Unit III – 10 Questions

UNIT I – Introduction

1. A McCulloch-Pitts neuron has inputs $x_1 = 1, x_2 = 0$, weights $w_1 = 2, w_2 = 1$, and threshold $\theta = 2$. What is the output?

A) 0
B) 1
C) 2
D) Cannot be determined

Answer: B) 1

2. Which of the following is the first artificial neuron model proposed in neural networks?

A) Perceptron
B) Sigmoid Neuron
C) McCulloch-Pitts Neuron
D) Adaline

Answer: C) McCulloch-Pitts Neuron

3. The primary limitation of a single-layer perceptron is that it cannot solve:

A) AND Gate
B) OR Gate
C) XOR Problem
D) NOT Gate

Answer: C) XOR Problem

4. The activation function used in the original Perceptron is:

A) Sigmoid Function
B) ReLU Function
C) Threshold (Step) Function
D) Tanh Function

Answer: C) Threshold (Step) Function

5. The output of the Sigmoid activation function always lies between:

A) -1 and 1
B) 0 and 1

- C) 0 and ∞
- D) $-\infty$ and $+\infty$

Answer: B) 0 and 1

6. A Perceptron has $w_1 = 1$, $w_2 = 2$, bias = -2 , and inputs $(1, 1)$. Using a step activation where output = 1 if net ≥ 0 , the output is:
- A) 0
 - B) 1
 - C) 2
 - D) -1

Answer: B) 1

7. In Gradient Descent, weights are updated in the direction of the:
- A) Positive Gradient
 - B) Negative Gradient
 - C) Random Direction
 - D) Zero Gradient

Answer: B) Negative Gradient

8. Which of the following statements is true about a Feed-forward Neural Network?
- A) Information flows backward only
 - B) Information flows in both directions
 - C) Information flows only from input layer to output layer
 - D) It contains feedback connections

Answer: C) Information flows only from input layer to output layer

9. Which of the following is the correct expression for the Sigmoid activation function?
- A) $f(x) = x^2$
 - B) $f(x) = \max(0, x)$
 - C) $f(x) = \frac{1}{1 + e^{-x}}$
 - D) $f(x) = \tan(x)$

Answer: C) $f(x) = \frac{1}{1 + e^{-x}}$

10. A major advantage of a Multi-Layer Perceptron (MLP) over a Single-Layer Perceptron is that it can:
- A) Solve only linear classification problems
 - B) Perform only regression tasks
 - C) Learn complex non-linear decision boundaries
 - D) Eliminate the need for training

Answer: C) Learn complex non-linear decision boundaries

11. The Perceptron Learning Algorithm is primarily used to:

- A) Reduce dimensionality
- B) Learn weights for a perceptron classifier
- C) Perform clustering
- D) Compute eigenvalues

Answer: B) Learn weights for a perceptron classifier

12. A perceptron can correctly classify data when the classes are:

- A) Randomly distributed
- B) Linearly separable
- C) Always nonlinear
- D) Unlabeled

Answer: B) Linearly separable

13. The net input to a neuron is generally computed as:

- A) $\sum w_i + x_i + b$
- B) $\sum w_i x_i + b$
- C) $\sum w_i / x_i$
- D) $\sum x_i - w_i$

Answer: B) $\sum w_i x_i + b$

14. The main purpose of a threshold function in a McCulloch-Pitts neuron is to:

- A) Normalize inputs
- B) Produce an output based on whether the weighted sum crosses a threshold
- C) Calculate eigenvalues
- D) Reduce the number of features

Answer: B) Produce an output based on whether the weighted sum crosses a threshold

15. Which type of neuron uses a smooth, differentiable activation function such as sigmoid?

- A) McCulloch-Pitts neuron only
- B) Sigmoid neuron
- C) Threshold neuron only
- D) Binary logic gate

Answer: B) Sigmoid neuron

16. If the learning rate in Gradient Descent is made excessively large, training may:

- A) Always converge faster

- B) Oscillate or diverge
- C) Stop requiring gradients
- D) Become independent of the loss function

Answer: B) Oscillate or diverge

17. The representation power of an MLP is mainly increased by introducing:

- A) Only more input features
- B) Hidden layers and nonlinear activations
- C) Only a larger learning rate
- D) PCA

Answer: B) Hidden layers and nonlinear activations

18. Which problem demonstrates the limitation of a single-layer perceptron?

- A) AND
- B) OR
- C) XOR
- D) NOT

Answer: C) XOR

19. A feed-forward neural network is characterized by:

- A) Cyclic connections
- B) Feedback loops
- C) Directed flow without feedback connections
- D) Recurrent connections

Answer: C) Directed flow without feedback connections

20. The primary role of an activation function in a neural network is to:

- A) Introduce non-linearity
- B) Store training data
- C) Calculate eigenvectors
- D) Reduce the dataset size

Answer: A) Introduce non-linearity

UNIT II – Feed Forward Neural Networks

1. A feed-forward neuron has $x_1 = 2, x_2 = 3, w_1 = 0.5, w_2 = 0.2, b = 0.2$ with linear activation. What is the output?
- A) 1.6
 - B) 1.8
 - C) 2.0
 - D) 2.2

Answer: B) 1.8

2. Weight update in Gradient Descent is performed in the direction of:
- A) +Gradient
 - B) –Gradient
 - C) Random
 - D) Zero

Answer: B) –Gradient

3. Momentum-based Gradient Descent uses the previous:
- A) Weight
 - B) Velocity
 - C) Bias
 - D) Epoch

Answer: B) Velocity

4. Nesterov Accelerated Gradient computes the gradient at the:
- A) Current position
 - B) Look-ahead position
 - C) Origin
 - D) Random point

Answer: B) Look-ahead position

5. AdaGrad mainly provides:
- A) Fixed learning rate
 - B) Adaptive learning rate
 - C) Momentum
 - D) Normalization

Answer: B) Adaptive learning rate

6. Adam combines ideas from:
- A) Momentum and RMSProp

- B) PCA and SVD
- C) SGD and PCA
- D) NAG and PCA

Answer: A) Momentum and RMSProp

7. If the product of eigenvalues of a matrix is -8 , and the eigenvalues are $k, k, -2$, the value of k is:

- A) 3
- B) 2
- C) -2
- D) -3

Answer: B) 2

8. The main objective of PCA is:

- A) Classification
- B) Dimensionality Reduction
- C) Regression
- D) Segmentation

Answer: B) Dimensionality Reduction

9. SVD decomposes a matrix A as:

- A) LU
- B) QR
- C) $U\Sigma V^T$
- D) LDL^T

Answer: C) $U\Sigma V^T$

10. Given $\eta = 0.1$, gradient = 4, and current weight = 3, the updated weight is:

- A) 2.6
- B) 3.4
- C) 2.4
- D) 2.9

Answer: A) 2.6

11. Backpropagation is primarily used to:

- A) Compute gradients of the loss with respect to network parameters
- B) Reduce image size
- C) Perform clustering
- D) Calculate PCA components

Answer: A) Compute gradients of the loss with respect to network parameters

12. Backpropagation relies heavily on which mathematical rule?

- A) Chain rule
- B) Bayes rule
- C) Product rule only
- D) Pythagorean theorem

Answer: A) Chain rule

13. In Stochastic Gradient Descent, a parameter update is typically calculated using:

- A) The entire dataset only
- B) A single example or small subset of examples
- C) The test set
- D) Eigenvectors

Answer: B) A single example or small subset of examples

14. RMSProp modifies the learning rate using:

- A) Accumulated squared gradients
- B) Class labels
- C) Eigenvectors
- D) Random noise

Answer: A) Accumulated squared gradients

15. Compared with standard Gradient Descent, Momentum mainly helps to:

- A) Accelerate movement in consistent directions
- B) Remove all gradients
- C) Reduce the dataset dimension
- D) Eliminate the loss function

Answer: A) Accelerate movement in consistent directions

16. An eigenvector v of matrix A satisfies:

- A) $Av = v$
- B) $Av = \lambda v$
- C) $Av = 0$ always
- D) $A + v = \lambda$

Answer: B) $Av = \lambda v$

17. The scalar λ in $Av = \lambda v$ is called the:

- A) Eigenvector
- B) Eigenvalue
- C) Singular vector

D) Bias

Answer: B) Eigenvalue

18. In PCA, the principal component corresponding to the largest eigenvalue represents the direction of:

- A) Minimum variance
- B) Maximum variance
- C) Zero variance
- D) Random variance

Answer: B) Maximum variance

19. Eigenvalue decomposition is commonly expressed in the form:

- A) $A = V\Lambda V^{-1}$
- B) $A = LU$
- C) $A = QR$
- D) $A = U + V$

Answer: A) $A = V\Lambda V^{-1}$

20. In SVD, Σ contains:

- A) Bias values
- B) Singular values
- C) Class labels
- D) Activation functions

Answer: B) Singular values

UNIT III – Autoencoders

1. An Autoencoder is primarily trained to:

- A) Classify data
- B) Reconstruct its input
- C) Predict class labels
- D) Generate random samples

Answer: B) Reconstruct its input

2. Which component of an Autoencoder produces the latent representation?

- A) Decoder
- B) Encoder
- C) Loss function
- D) Optimizer

Answer: B) Encoder

3. An Autoencoder is called **Undercomplete** when:

- A) $d > n$
- B) $d = n$
- C) $d < n$
- D) $d = 0$

Answer: C) $d < n$

4. Which loss function is commonly used for reconstruction of real-valued inputs?

- A) Mean Squared Error
- B) Hinge Loss
- C) Accuracy
- D) F1 Loss

Answer: A) Mean Squared Error

5. A Linear Autoencoder with linear encoder, linear decoder, MSE loss, and centered data is equivalent to:

- A) K-Means
- B) PCA
- C) SVM
- D) Decision Tree

Answer: B) PCA

6. A Denoising Autoencoder is trained using:

- A) Clean input and noisy target

- B) Noisy input and clean target
- C) Noisy input and noisy target only
- D) Labels and noisy input

Answer: B) Noisy input and clean target

7. The main objective of a Sparse Autoencoder is to:

- A) Increase the number of active neurons
- B) Encourage sparse hidden representations
- C) Remove the decoder
- D) Increase reconstruction error

Answer: B) Encourage sparse hidden representations

8. A Contractive Autoencoder primarily encourages the learned representation to be:

- A) Highly sensitive to small input changes
- B) Insensitive to small changes in the input
- C) Completely random
- D) Independent of the input

Answer: B) Insensitive to small changes in the input

9. Which regularization technique randomly sets a fraction of neuron outputs to zero during training?

- A) Batch Normalization
- B) Early Stopping
- C) Dropout
- D) PCA

Answer: C) Dropout

10. The main purpose of L2 regularization in a neural network is to:

- A) Increase the magnitude of weights
- B) Penalize large weights
- C) Remove the decoder
- D) Increase the learning rate

Answer: B) Penalize large weights

DEEP LEARNING

MID-1 – Fill in the Blanks Question Bank

Unit I: 10 Questions Unit II: 10 Questions Unit III: 5 Questions

UNIT I – Introduction

1. The first artificial neuron model proposed in neural networks was the _____ neuron.

Answer: McCulloch-Pitts

2. A McCulloch-Pitts neuron uses a _____ activation function to produce its output.

Answer: Threshold (Step)

3. The primary limitation of a single-layer Perceptron is that it cannot solve the _____ problem.

Answer: XOR

4. The Perceptron Learning Algorithm is used to learn the _____ of a Perceptron.

Answer: Weights

5. A single-layer Perceptron can correctly classify data when the classes are _____ separable.

Answer: Linearly

6. The activation function of a Sigmoid neuron is given by $f(x) = \frac{1}{1 + e^{-x}}$.

Answer: $\frac{1}{1 + e^{-x}}$

7. The output of the Sigmoid activation function lies between _____ and _____.

Answer: 0 and 1

8. In Gradient Descent, the weights are updated in the direction opposite to the _____.

Answer: Gradient

9. A Multi-Layer Perceptron can learn complex _____ decision boundaries.

Answer: Non-linear

10. In a Feed-forward Neural Network, information flows from the _____ layer towards the output layer without feedback connections.

Answer: Input

UNIT II – Feed Forward Neural Networks

1. The algorithm used to compute gradients of the loss with respect to the parameters of a multilayer neural network is called _____.

Answer: Backpropagation

2. Backpropagation is based on the mathematical _____ rule.

Answer: Chain

3. Momentum-based Gradient Descent uses the previous _____ to accelerate optimization.

Answer: Velocity

4. Nesterov Accelerated Gradient computes the gradient at the _____ position.

Answer: Look-ahead

5. In Stochastic Gradient Descent, the model parameters are updated using a single training example or a small _____.

Answer: Mini-batch

6. AdaGrad mainly provides an _____ learning rate for different parameters.

Answer: Adaptive

7. RMSProp uses an exponentially moving average of _____ gradients to adapt the learning rate.

Answer: Squared

8. Adam combines the ideas of Momentum and _____.

Answer: RMSProp

9. In the eigenvalue equation $Av = \lambda v$, the scalar λ is called the _____.

Answer: Eigenvalue

10. SVD decomposes a matrix A into the form $A =$ _____.

Answer: $U\Sigma V^T$

UNIT III – Autoencoders

Note: Unit III questions are restricted to topics up to Contractive Autoencoders for MID-1.

1. An Autoencoder is primarily trained to _____ its input.

Answer: Reconstruct

2. The component of an Autoencoder that produces the latent representation is called the _____.

Answer: Encoder

3. An Autoencoder is called undercomplete when the latent dimension d is _____ the input dimension n .

Answer: Less than ($d < n$)

4. A Denoising Autoencoder receives a _____ input and tries to reconstruct the clean input.

Answer: Noisy

5. A Contractive Autoencoder encourages the learned representation to be insensitive to small _____ in the input.

Answer: Changes